

$$0 + 1 + 0 + 2 = \frac{3}{10}$$

CMI
BVR

Probability
Quiz 1

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If you are using a theorem, quote it precisely. Simplify your answer.

1. In Bose-Einstein experiment with 30 balls and 5 boxes what is the most probable number of balls in Box 2? Justify your answer.

↳ 30 balls, 5 boxes
not unique?

you did not understand the problem [3]

Let X_i = number of balls in Box 2, want $E X$

$$X = 0, 1, 2, \dots, 30$$

$$P X =$$

$$\binom{30}{i}$$

$$E X = \sum_{R=0}^{30} R \frac{\binom{30}{R}}{5^{30}} = \frac{1}{5^{30}} \sum_{R=0}^{30} R \binom{30}{R}$$

$$= \frac{1}{5^{30}} \cdot 30 \cdot 5^{29} = 30 \cdot \frac{1}{5} = 6$$

$$\therefore E X = 6$$

guess:
equal distribution means
→ 6 → ✓?

Rough:
suppose
exactly 2 balls in box 2

$$\binom{30}{2}$$

$$(1+x)^{30} = \binom{30}{0} + \binom{30}{1}x + \dots$$

If Bose-Einstein
to some balls, then
1... standard Bose
= $\binom{n+r-1}{r-1}$?

↳ Then get for box 2?

2. A certain component in a rocket engine fails 5 percent of the time when the engine is fired. To achieve greater reliability in the engine working, this component is duplicated n -times. The engine then fails only if all of these n components fail. Assume the component failures are independent of each other. What is the smallest value of n that can be used to guarantee that the engine works 99.999 percent of the time? [3]

$$P(\text{Failure}) = \frac{5}{100} \text{ or } \frac{1}{20}$$

For engine to fail, all components must fail,

$$\text{so } P(\text{all fail}) = \left(\frac{1}{20}\right)^n$$

$$\text{so } P(\text{at least one succeeds}) = 1 - P(\text{all fail}) = 1 - \left(\frac{1}{20}\right)^n$$

↳ We want this to be $\geq \frac{99.999}{100}$

$$\text{so } 1 - \left(\frac{1}{20}\right)^n \geq \frac{99.999}{100}$$

$$\text{so } \frac{100 - 99.999}{100} \geq \frac{1}{20^n} \rightarrow \frac{0.001}{100} \geq \frac{1}{20^n}$$

$$\rightarrow 20^n \geq \frac{100}{0.001} = 100,000$$

$$\text{so } n \geq \log_{20}(100,000)$$

↳ choose smallest natural number bigger than this.

Should we look at logarithmic tables?
This is high school problem

3. Three independent events occur with probabilities $1/4$, $1/3$ and $1/2$ respectively. Given that at least two of the three events occurred what is the probability that the second event did not occur? [2]

→ let events be A, B, C

→ $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{3}$, $P(C) = \frac{1}{2}$

Want → $P(B^c | \text{at least 2 of the events occurred})$
 $= 1 - P(B | \text{at least 2 of the events occurred})$

$= 1 - (P(B|A \cap C) + P(B|B \cap C) + P(B|A \cap B) - P(B|A \cap B \cap C))$? No.

$= 1 - \left(\frac{P(B \cap A \cap C)}{P(A \cap C)} + \frac{P(B \cap B \cap C)}{P(B \cap C)} + \frac{P(B \cap A \cap B)}{P(A \cap B)} - \frac{P(B \cap A \cap B \cap C)}{P(A \cap B \cap C)} \right)$

$= 1 - (P(B) + 1 + 1 - 1) \rightarrow \text{No!}$

→ at least 2 of 3 events occur

→ A occurs or B occurs or C occurs, or A and B, or A and C, or B and C

2nd did not occur: (C or A) also

$= \frac{P(A \cap C)}{P(A \cap B) + P(B \cap C) + P(A \cap C) - P(A \cap B \cap C)}$?

→ (as they already considered!)

$= \frac{\frac{1}{4} \times \frac{1}{2}}{\frac{1}{12} + \frac{1}{6} + \frac{1}{8} - \frac{1}{24}} = \frac{\frac{3}{24}}{\frac{2}{24} + \frac{4}{24} + \frac{3}{24} - \frac{1}{24}}$

$P(B^c | \text{given at least 2 of the events occurred}) = \frac{3}{8}$

Too much English is distracting you - stick to definitions

You can do.

4. If $X \sim G(p)$, find $E[\sin(\pi X/2)]$.

[2]

$$\hookrightarrow X \sim G(p)$$

$$\text{so } P(X=k) = q^k p, \text{ where } q = (1-p)$$

$$\text{We know, } E[f(X)] = \sum f(x_i) p_i$$

And X takes values $0, 1, 2, \dots$

$$E(\sin(\pi X/2)) = \sum_{i=0}^{\infty} \sin\left(\frac{\pi \cdot i}{2}\right) \cdot p_i$$

$$= \underbrace{\sin(0)}_0 p_0 + \underbrace{\sin\left(\frac{\pi}{2}\right)}_1 p_1 + \underbrace{\sin(\pi)}_0 p_2 + \underbrace{\sin\left(\frac{3\pi}{2}\right)}_{-1} p_3 + \dots$$

$$= p_1 - p_3 + p_5 - p_7 + \dots$$

$$= q^1 p - q^3 p + q^5 p - q^7 p + \dots$$

$$= p(q - q^3 + q^5 - q^7 + \dots)$$

$(-q^2)$ multiplied each time

$$\frac{q((q^2)^n - 1)}{(q^2)^n - 1} = \frac{q}{1+q^2}$$

$$E\left[\sin\left(\frac{\pi X}{2}\right)\right] = \frac{pq}{1+q^2}$$